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Flapping-Wings Drones for Pests and Diseases Detection in Horticulture

Orel Awesta, Vera Hollink, and Mauro Gallo

***Abstract*—Agriculture and horticulture are essential for ensuring safe food to the growing global population, but they also contribute significantly to climate change and biodiversity loss due to the extensive use of chemicals. Integrated pest management is currently employed to monitor and control pest populations, but it relies on labor-intensive methods with low accuracy. Automating crop monitoring using aerial robotics, such as flapping-wing drones, presents a viable solution. This study explores the application of deep learning algorithms, You Only Look Once (YOLO) and Faster region-based convolutional neural network regions with convolutional neural networks (R-CNN), for pest and disease detection in greenhouse environments. The research involved collecting and annotating a diverse dataset of images and videos of common pests and diseases affecting tomatoes, bell peppers, and cucumbers cultivated in Dutch greenhouses. Data augmentation and image resizing techniques were applied to enhance the dataset. The study compared the performance of YOLO and Faster R-CNN, with YOLO demonstrating superior performance. Testing on data acquired by flapping-wing drones showed that YOLO could detect powdery mildew with accuracy ranging from 0.29 to 0.61 despite the shaking movement induced by the actuation system of the drone's flapping wings.**

***Index Terms*—Deep learning (DL), flapping-wing drones (FWDs), greenhouse crops, horticulture, integrated pest management (IPM), pest and disease detection.**

I. Introduction

AGRICULTURE and horticulture are pivotal in ensuring the provision of safe food for the growing global population [1]. While these sectors are indispensable for sustaining human life by providing essential food resources, they also significantly contribute to climate change and biodiversity loss. This is primarily due to the extensive use of chemicals, such as pesticides, fungicides, and fertilizers, employed to safeguard crops from pests and diseases. Although the application of pesticides enables farmers and growers to protect their crops and secure profits, it simultaneously degrades food quality and poses health risks. Consequently, policymakers are enacting increasingly stringent regulations aimed at drastically reducing pesticide usage in agriculture and horticulture [2].

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Fig. 1. Exemplary of yellow sticky cards in greenhouse crops (source: web).

Currently, greenhouse growers employ the integrated pest management (IPM) methodology to monitor and control pest populations and diseases. Based on the data acquired, they can make informed decisions regarding the strategies to mitigate crop damage caused by pests and diseases [3]. A critical component of the IPM is monitoring. Accurate and reliable monitoring of greenhouse plants is essential for detecting and identifying pests and diseases, as well as quantifying their spread within the crops. Presently, this IPM operation is conducted using yellow sticky cards that attract and capture pests (see Fig. 1). These cards necessitate regular inspections by operators to identify pest species and estimate their distribution. However, this monitoring method is characterized by low accuracy and does not provide precise localization of pests within the greenhouse. In addition, it is inadequate for detecting fungi and diseases. The reliance on human labor for monitoring is problematic due to the global labor shortage affecting the agriculture and horticulture sectors [4].

Automating crop monitoring offers a promising solution to these challenges. By enabling early pest detection, automated systems allow growers to address pest issues using more sustainable methods, such as biopesticides and natural predators. These remedies are effective only when pests or infections are identified at an early stage. Furthermore, automating crop monitoring can potentially alleviate the labor shortages in agriculture and horticulture. While robotic arms can be used for plant scouting, they are expensive and necessitate the installation of costly infrastructure in greenhouses. An alternative approach involves the use of aerial robotics. At the Inholland University of Applied Sciences, we have pioneered the application of propeller drones in greenhouses as part of the HiPerGreen project [5]. However, quadcopters, due to the damage caused by high-speed propellers and strong downwash, are suitable only for ornamental crops,

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Fig. 2. Example of the FWD (courtesy Flapper drones).

where plants are horizontally distributed in the greenhouse. For vertically growing plants, such as tomatoes and bell peppers, a bioinspired aerial robotic solution is more appropriate. Drones with flapping wings (see Fig. 2), inspired by nature, have proven to be more plant friendly [6]. Flapping-wing drones (FWD), or ornithopters, are bioinspired aerial vehicles designed to mimic the flight mechanisms of birds and insects. Their flight relies on the dynamic motion of flexible wings, offering advantages over traditional fixed- and rotary-wing drones in terms of energy efficiency and maneuverability. FWDs achieve exceptional maneuverability through asymmetric wing strokes and independent wing articulation. These capabilities allow them to make sharp turns, hover, and transition between different flight modes efficiently making them ideal for navigating confined spaces, such as the intricate foliage of horticulture high wire crops [7]. The development of FWDs presents significant engineering challenges, including material selection, actuation mechanisms, and control system integration. Lightweight yet durable materials, such as carbon fiber and polymer films, are essential for achieving optimal wing flexibility and strength. Furthermore, the precise actuation of wings requires efficient mechanisms, such as servos or piezoelectric actuators, to replicate natural wing movements while minimizing energy consumption. These drones, similar to the quadcopters, can be equipped with sensors and cameras to collect data from the crops and measure environmental conditions.

In this article, we present a solution for pest and disease recognition utilizing images captured by cameras mounted on FWDs. This approach aims to improve early pest detection, thereby promoting the use of sustainable pest control methods and reducing the horticulture sector reliance on human labor.

The rest of this article is organized as follows. Section II provides an overview of the current state of the art in deep learning (DL) applications for pest and disease monitoring in agriculture and horticulture. Section III describes how the datasets used for training the DL models were obtained and details the data preprocessing procedures. Section IV provides a description of the selected DL models and a discussion of the results. Finally, Section V concludes this article and outlines recommendations for future work.

II. Related Work

DL is a branch of machine learning that uses neural networks with many layers to model complex patterns in data. These neural networks are inspired by the human brain structure and function. DL algorithms can automatically learn and improve from large amounts of data, making them particularly effective for tasks like image and speech recognition, natural language processing, and more. DL techniques have proven effective in pest and disease detection by using data from images and videos. Wanga et al. [8] introduced Pest24, a comprehensive big data platform designed for intelligent multipest monitoring in agriculture. Fuentes et al. [9] explored various deep learning architectures to determine the most suitable ones for detecting pests and diseases in tomato crops. Xia et al. [10] developed a pest identification method using yellow sticky traps, the watershed algorithm, and Mahalanobis distance in the YCrCb color space. This method is both computationally efficient and highly effective in identifying common greenhouse pests, such as aphids and thrips.

Arumuga and Umamaheswari [11] evaluated four DL object detection models (single shot multibox detector (SSD), efficient DETection (EfficientDET), Faster region-based convolutional neural network (R-CNN), and CenterNet) with five backbone networks (MobileNet V2, ResNet101 V1, Inception ResNet V2, EfficientNet, and HourGlass104). Their study revealed that Faster R-CNN ResNet101 V1 outperformed the other models, achieving a mean average precision (mAP) of 74.77% . Coulibaly et al. [12] addressed the challenge of DL models often being used as closed boxes, which complicates understanding their learning processes. They enhanced model explainability through visualization techniques, improving user interaction. Teng et al. [13] identified that poor performance in pest recognition with DL is often due to neglecting object distribution and position requirements. They proposed a robust pest detection network featuring a multiscale superresolution feature enhancement module and a soft intersection-over-union (IoU) mechanism. Their approach achieved a 67.4% mAP on the LLPD-26 dataset, outperforming state-of-the-art models attention-guided faster R-CNN (AF-RCNN) and hierarchical global-local attention (HGLA) by 15.0% and 2.7%, respectively.

The utilization of aerial robotics (drones) for crop monitoring is predominantly implemented in open-field agriculture. The application of drones for crop monitoring within greenhouses remains relatively limited. This limitation arises from the lack of GPS signal coverage within greenhouses, which poses significant challenges for the autonomous navigation of drones due to the difficulty in determining their position.

Few Dutch companies and startups, such as Pats drones, Corvus, and Avular, are developing drone technology for various sectors, including horticulture. Their applications primarily focus on pest control and crop monitoring and counting, particularly in ornamental crops where plants are horizontally distributed within greenhouses. Autonomous navigation in these environments is facilitated by the placement of markers (QR codes) within the greenhouse.

This article presents the research contributions to the Flappy project [14]. The objective of this project is to develop and implement a drone-based monitoring system for horticulture. To achieve this, the present study opted to train DL models to detect the most common pests and diseases affecting crops in the Dutch horticultural landscape. This study represents a first

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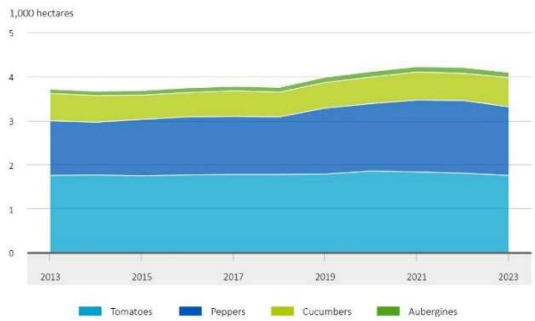

Fig. 3. Popular vegetables grown in the Dutch greenhouses [15].

attempt to validate DL algorithms using real data (videos and images) acquired by FWDs (see Fig. 2).

III. Data Production and Acquisition

The most common horticultural plants in The Netherlands are tomatoes, bell peppers, and cucumbers (see Fig. 3).

Accordingly, the collection and generation of data are focused on specific types of pests and diseases commonly affecting and generally prevailing on these plants. These pests and diseases, included in our dataset, concern leaf miner, aphids, Tuta absoluta, and powdery mildew. The dataset covers a variety of manifestations of pests and diseases, ranging from initial symptoms to advanced stages of infestation. These stages reflect different levels of severity, with advanced stages potentially causing significant crop damage. This variability is crucial for training models to identify pests and diseases across different points in their life cycles. The dataset was collected in various sessions and stages. The first set of images (1597 images) was obtained from various greenhouses and crops, specifically cucumbers, bell peppers, and tomatoes. This collection aimed to document diverse environmental conditions within the greenhouses, including variations in lighting, plant species, and the presence of pests such as aphids, powdery mildew, leaf miners, and Tuta absoluta (see Fig. 4).

A second set of approximately 2000 images was collected from a specialized greenhouse compartment, established in collaboration with the research company Vertify at the World Horti Center in Naaldwijk, The Netherlands (see Fig. 5). During a three-month period, this compartment was used to cultivate cucumbers that were deliberately infected with aphids and powdery mildew. Images were captured periodically to monitor the developmental stages of these pests and diseases, thereby ensuring the availability of images of all developmental stages.

Images were acquired using different types of smartphone cameras with an image resolution of 12 megapixels. In addition, a series of test campaigns was conducted to collect data concerning pests and diseases, using FWDs. During these experimental trials, the FWD was operated remotely, capturing both video and image data through a camera. A custom 3D-printed mount was fabricated to secure the camera to the drone. The camera utilized (see Fig. 6) is the Insta360 GO 3S, which offers a 12-megapixel image resolution and 4K video resolution. It features a 360° field of view and is powered by a 310-mAh battery, providing a maximum operational duration of 38 min. The camera dimensions are 2.6 × 5.4 × 3.2 cm, and it has a mass of approximately 32 g. The flight duration was significantly influenced by the additional weight of the installed camera used for the crop monitoring. The drone is powered with a LiPo 2S 300-mAh battery providing an estimated flight time of approximately 8 min without any payload and for a forward velocity of about 3 m/s. However, the inclusion of the camera for the pest

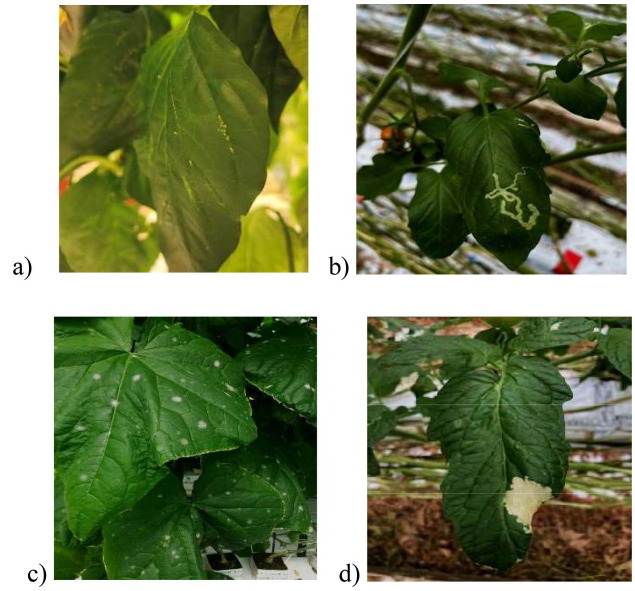

Fig. 4. Data example: (a) Aphids, (b) leaf miner, (c) powdery mildew, and (d) Tuta absoluta.

Fig. 5. Greenhouse compartment (cucumber plants).

Fig. 6. Camera on the FWD.

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detection reduced the flight duration to 2 min. To mitigate further reductions in flight time, a horizontal scanning approach was adopted, and the flight speed was minimized. For each trial, the FWD scanned a crop length of about 100 m while maintaining a constant flight altitude. The distance from the plants varied between 20 and 100 cm, depending on environmental factors such as the presence of leaves, branches, and other obstacles. FWDs can generate vibrations that negatively impact video quality. One potential solution involves hardware stabilization using a gimbal; however, this option was not implemented due to payload constraints. Alternatively, the adverse effects of vibrations can be mitigated through postprocessing by applying various video stabilization algorithms to the acquired data. In this study, the vibrational effects induced by the flapping wings have been mitigated through the activation of the FlowState stabilization technology integrated into the selected camera, the Insta360 GO 3s.

A. Data Preprocessing

Data preprocessing transforms raw data to prepare it for neural network consumption. This step is crucial for effective model training, validation, and deployment in object detection projects. It enhances model quality, reliability, and generalization, resulting in more accurate and robust detection outcomes [16]. The adopted data preprocessing consists of two operations: data augmentation and image resizing and scaling, as detailed in the next sections. As previously mentioned, our final dataset, which combines the first and second datasets, includes images captured under diverse conditions, simulating real-world scenarios. This variability ensures that our model is trained to manage different lighting conditions, angles, and distances, thereby enhancing its ability to filter out noise and distractions and focus on detection (identifying the presence of pests and diseases), localization (pinpointing the exact position of the pest/disease within the image), and classification (categorizing the pests and diseases into predefined classes). The dataset was annotated using Roboflow [17], a free and user-friendly tool for managing image datasets, by manually drawing bounding boxes around pests in each image and labeling them according to their respective categories. Subsequently, the dataset was divided into three parts: 70% for the training set, which is used to teach the model to identify, locate, and classify our object; 20% for the validation set, which is used to fine-tune the model hyperparameters and prevent overfitting; and 10% for the test set, which is used to assess the model performance on new unseen data.

B. Data Augmentation

Data augmentation is a technique used to artificially increase the diversity of a dataset by applying various transformations to the images, with the aim of enhancing the model ability to generalize object features and improving its accuracy in pest and disease detection [18]. The data augmentation was applied to the original dataset prior to DL model training. The augmentation techniques employed include horizontal and vertical flips, random saturation adjustments (ranging from –30% to 30%), random brightness adjustments (ranging from –20% to 20%), and exposure adjustments (ranging from –11% to 11%). These

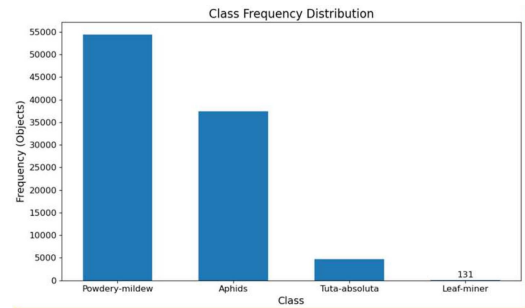

Fig. 7. Class distribution in the dataset.

techniques are exclusively applied to the training dataset to simulate real-world scenarios where a drone captures images of pests and diseases. Each image in the training dataset is augmented three times using these techniques, resulting in a training set that has three times as many images, thereby increasing both the size and diversity of the dataset.

C. Image Resizing and Scaling

The dataset comprises high-resolution images (3024 × 4032 pixels), which necessitate resizing and scaling to meet the specific dimensional requirements of neural network models. This standardization ensures compatibility with our model architecture. The selected algorithms, YOLOv9 and Faster R-CNN (see Section IV), require minimum image sizes of 640 × 640 and 600 × 600 pixels, respectively. Consequently, we resized the dataset images to 640 × 640 pixels. Although larger input sizes require more computational resources, this resizing facilitates a more consistent comparative analysis of You Only Look Once (YOLO) and Faster R-CNN performance. In addition, the "Auto-Orient" function was used to automatically adjust the orientation of images based on their embedded metadata [19]. This metadata, often generated by cameras or imaging devices, can indicate how an image should be displayed (e.g., rotated or flipped). By applying "Auto-Orient," the images are physically reoriented to match the intended display orientation, ensuring consistent appearance across all images, regardless of the original metadata. After preprocessing, the final training set comprises 5325 images, the validation set includes 508 images, and the testing set consists of 253 images.

D. Preliminary Exploratory Data Analysis

A preliminary exploratory data analysis provides key insights into the dataset features critical for evaluating model performance. The first aspect to analyze is the class distribution, which refers to the frequency of samples in each category or class within the dataset. Imbalances caused by over- or under-represented classes can hinder equitable learning and lead to biased results. Conversely, a balanced representation of classes is crucial for achieving optimal model performance across all categories [20], [7]. As expected, annotations in the dataset reveal an imbalance, with powdery mildew and aphids being more prevalent than other pests, as illustrated in Fig. 7.

Smaller pests, such as aphids, are more numerous, whereas larger pests occur less frequently. These differences in size and prevalence underscore the importance of considering both size

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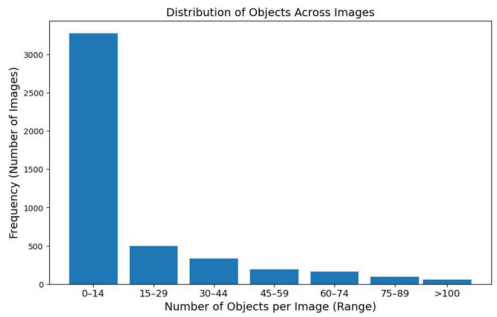

Fig. 8. Object distribution over the image dataset.

and frequency of the pests when evaluating class distribution. Therefore, despite the class imbalance, we have decided to proceed with the current dataset. This is because although Tuta absoluta and leaf miner are underrepresented, we anticipate that the model will perform reasonably well on these classes due to their relatively large size and distinct structural features, which make them easier to identify compared to smaller pests such as aphids. Nevertheless, we will take the class distribution into account when evaluating the models performance.

Image resolution is a critical aspect that significantly influences model performance and generalization [21]. Standardizing the resolution across all images in our dataset ensures consistent inputs for the object detection models. Moreover, detecting small pests, such as aphids, requires high-resolution images to preserve fine details. This, in turn, influences the choice of algorithms, favoring those that are compatible with larger image sizes to ensure the accurate detection of tiny objects. This approach enhances the precision of detecting small objects. The distribution of pests and diseases across images in our dataset provides insights into the complexity and diversity of the images. Knowing the number of objects per image helps in understanding the level of detail and the challenges in detecting small or densely packed objects. For this analysis, an object distribution graph was generated from our dataset (see Fig. 8). According to this graph, the object distribution in the dataset highlights variability, with most images containing 0–14 objects and a few cases containing more than 100 objects. Considering that pests like aphids and diseases such as powdery mildew were annotated individually, this distribution reflects the clustering behavior of both small pests, like aphids, and fungal diseases, like powdery mildew.

The last aspect concerns background complexity and color discrepancy of the images, which further impact the detection of the objects, influencing the model performance [22]. Green pests, such as aphids, often blend into dark green foliage, making it necessary to apply preprocessing techniques to improve contrast and visibility. Lighting conditions also affect detection, with images taken under natural daylight providing better visibility of the pests compared to artificial lighting.

IV. DL Models Selection: Training, Validation, and Testing Framework

DL, a subfield of machine learning, mimics human cognitive processes and specializes in managing complex tasks, particularly in the domains of audio, image, and video recognition. These applications require the identification of numerous intricate input features, highlighting the critical importance of neural networks [21]. Considering greenhouse application requirements, computational resources, required input size, and the tradeoff between speed and accuracy, we compare YOLOv9 and Faster R-CNN. The models represent the two main categories of object detection models: one- and two-stage models [13]. For real-time detection scenarios such as field crop inspection, efficient one-stage algorithms like YOLO are preferred [23], [24]. Conversely, for the precise detection of pests and diseases, the Faster R-CNN model is deemed most suitable [25]. Therefore, in Section V, we will evaluate the performance of a one-stage network (YOLOv9) against a two-stage network (Faster R-CNN) using our current dataset.

A. One-Stage Neural Networks (YOLO)

One-stage neural network algorithms have become prominent due to their simplicity and efficiency in object detection, achieved by integrating region proposals and classification stages into a single step. This integration reduces computational complexity and inference time [13], thereby enabling real-time object detection even on resource-constrained devices. These algorithms predict both object classes and bounding boxes in a single forward pass through the network, underscoring their suitability for a wide range of object detection applications. YOLO, introduced by Redmon et al. [26], is a prominent one-stage neural network renowned for its real-time performance, achieved by simultaneously predicting bounding boxes and class probabilities. Its simplicity and speed render it ideal for rapid pathogen detection in applications, such as agricultural monitoring. The requirement of 640 $\times$ 640 pixels for input images enhances YOLO effectiveness in detecting very small objects, including pests. YOLOv9, introduced in February 2024, incorporates a novel approach to mitigating information loss [27]. Although untested on tasks similar to ours, this state-of-the-art version merits exploration in our research due to its potential to detect, localize, and classify small objects like pests and diseases in greenhouse crops.

B. Two-Stages Neural Networks (Faster R-CNN)

Two-stage network algorithms prioritize accuracy and robustness in object detection, distinguishing them from their one-stage counterparts. These algorithms initially generate region proposals, which are candidate regions in an image likely to contain objects, using methods such as selective search or region proposal networks (RPN), followed by the classification and refinement of these proposals [20]. Notable examples include the R-CNN and its variants, such as Fast R-CNN, Faster R-CNN, and Mask R-CNN. Despite their slower processing speed, we selected the Faster R-CNN for its renowned accuracy, precise localization, and robust classification capabilities, which align perfectly with the demanding requirements of our task. Deploying two neural network algorithms with distinct functionalities allows us to assess the feasibility of detecting, localizing, and classifying pests and diseases.

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C. Evaluation Measures

To train and test the two algorithms, we utilized statistical measures, such as average precision (AP), average recall (AR), and mAP. AP is calculated as follows:

$$\mathrm{AP} = \int_0^1 p(r)\, dr \tag{1}$$

where *p(r)* is the precision as a function of the recall *r*

$$\begin{aligned} p &= \frac{\text{Correct predictions}}{\text{Total predictions}} \\ &= \frac{\text{True positive (TP)}}{\text{True positive (TP)} + \text{False positive (FP)}} \end{aligned} \tag{2}$$

$$\begin{aligned} r &= \frac{\text{Correct predictions}}{\text{Total ground truth}} \\ &= \frac{\text{True positive (TP)}}{\text{True positive (TP)} + \text{False negative (FN)}}. \end{aligned} \tag{3}$$

In object detection, AP (1), ranges from 0 to 1, is a metric used to evaluate model performance as it provides a comprehensive measure of a model ability to correctly identify and classify objects at varying confidence thresholds, with higher AP values indicating superior detection accuracy [28].

AR is calculated as follows:

$$\mathrm{AR} = \int_{0.5}^1 r(t)\, dt, \quad t \in [0.5,\ 1] \tag{4}$$

where *r(t)* is the recall as a function of the overlap *t*, measured by IoU, and refers to the ratio of the intersection area between the predicted bounding box and the ground truth box to their combined area. In object detection, AR is a performance metric that summarizes the recall values across different IoU thresholds. It reflects the model ability to identify all relevant objects in a given dataset, with higher AR values indicating a greater capacity to detect true positives across varying levels of detection confidence.

mAP is expressed as

$$\mathrm{mAP} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{AP}_i \tag{5}$$

where *N* is the number of classes. mAP (5) calculates the AP and recall scores for each object class to determine the overall accuracy of the object detector. mAP50 measures accuracy with a 50% overlap (IoU 0.50), mAP95 with a 95% overlap requirement (IoU 0.95), and mAP50:95 averages precision across multiple overlap thresholds from 50% to 95%. In this research, we use mAP50, mAP50:95, and AR to evaluate the performance of our models, as they are the most commonly used metrics for assessing object detection performance [29].

D. Hyperparameter Tuning

During neural network training, several hyperparameters are crucial for training object detection models, as they directly impact the model ability to learn effectively, generalize well, and perform accurately on unseen data. In our training, we focused on the batch size (BS), epochs (Ep), image size (Im size), and learning rate (Lr), as defined shortly. Proper tuning of these parameters maximizes model performance for our detection task [30]. *Batch size*, the number of training examples processed in one iteration, impacts computational requirements and model convergence. *Epochs* denote a complete pass of the training dataset through the neural network, updating model parameters based on the training data loss. *Image size* specifies the dimensions of input images loaded into the model during training, affecting the network ability to learn from visual data. The *learning rate* governs the size of steps taken during optimization to adjust weights and biases. Its selection significantly impacts convergence speed and training quality. A high learning rate may cause optimization to oscillate or fail to converge, while a low learning rate can slow down the process. Identifying the right balance by choosing an optimal learning rate is vital for achieving the best performance from the neural network.

TABLE I
YOLO HYPERPARAMETER OPTIMIZATION

Hyperparameters				Evaluation measures		
Im size	**BS**	**Ep**	**Lr**	**mAp50**	**mAp50:95**	**AR**
640×640	8	50	0.001	46%	30%	49%
640×640	16	50	0.01	48.80%	30.3%	49.3%
850×850	8	50	0.01	**50%**	**31%**	46%
640×640	4	50	0.01	42.3%	27.6%	41%

TABLE II
YOLO CLASS PERFORMANCE AFTER HYPERPARAMETER TUNING

Class	mAP50	mAP50:95	AR
Aphids	9%	2.7%	8.3%
Tuta absoluta	70%	46.1%	66%
Powdery mildew	51.6%	22%	52.7%
Leaf miner	69%	53.6%	57.1%
Average total	50%	31%	46%

E. One-Stage Object Detection: YOLOv9

In this study, we opted for the compact model YOLOv9-C (see Appendix I for the model architecture). Subsequently, multiple experiments were conducted to identify optimal parameters, as detailed in Table I. The highest mAP50 and mAP50:95 achieved were 50% and 31%, respectively, with an image input size of 850 × 850. The AR, on the other hand, seems to be higher by the trials with the smaller input size (640 × 640).

Upon completing the training of the YOLO model, various analysis graphs are generated to depict cross-entropy loss, detection accuracy, and classification accuracy as functions of the number of epochs (see Appendix II).

Table II presents the model performance on a per-class basis, where the mAP score (mAP50), with an IoU threshold of 0.5 across all classes, is 50%. This result indicates that the model

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TABLE III
FASTER R-CNN HYPERPARAMETER TUNING

Hyperparameters				Evaluation measures		
Im Size	**BS**	**Ep**	**Lr**	**mAP50**	**mAP50:95**	**AR**
640 × 640	16	30	0.001	37.09%	21.38%	27.6%
640 × 640	8	30	0.01	**46.81%**	**28.42%**	**34.70%**
850 × 850	8	30	0.001	43.6%	26.5%	30.3%
640 × 640	4	30	0.01	35.6%	20.23%	25.13%

See Section IV-D for the explanation of the abbreviations BS, Ep, Im size, and Lr.

TABLE IV
FASTER R-CNN CLASS PERFORMANCE AFTER HYPERPARAMETER TUNING

Class	mAP50:95	AR50:95
Aphids	6%	11.5%
Tuta absoluta	48.7%	58%
Powdery mildew	17.5%	26.3%
Leaf miner	39.4%	43%
Average total	28%	34.7%

accurately identifies pests or diseases approximately 50% of the time with reasonable accuracy and recall. Given the imbalanced class distribution discussed previously, it is noteworthy that despite the underrepresentation of the leaf miner and Tuta absoluta classes, the model demonstrates relatively strong detection performance for these categories, with mAP50 values of 69% and 70%, respectively, and high AR scores (57.1% and 66%). In contrast, the detection of aphids poses a significant challenge, as these pests are particularly difficult to observe, even with the naked eye. This difficulty is reflected in the notably low mAP50 (9%) and AR (8.3%) for Aphids.

F. Two-Stage Object Detection: Faster R-CNN

For our research, we have utilized the Faster R-CNN pipeline repository, which is compatible with various backbones and is designed to simplify training (see Appendix III for the model architecture) [31]. Consequently, we conducted hyperparameters tuning by adjusting the image size (Im size), learning rate (Lr) and the batch size (BS) (see Table III). With these refined parameters, the model with an image size of 640 × 640 demonstrated superior performance, achieving mAP50 and mAP50:95 scores of 46.81% and 28.42%, respectively. This indicates that, on average, the model achieves 46.81% precision for detecting objects with at least 50% overlap with the ground truth (mAP50) and 28.42% precision when evaluated across IoU thresholds ranging from 50% to 95% in steps of 5% (mAP50:95).

Also here, upon completing the training of the Faster R-CNN model, various analysis graphs are generated to depict training and validation loss as functions of the number of steps, which represent the iterations of gradient updates during training (see Appendix IV). Table IV shows the Faster R-CNN model performance on a per-class basis. The highest mAP score (mAP50:95) across all classes is 28%, with an AR (AR50:95) of 34.7%. While the model overall struggles with detecting aphids (mAP50:95 of 6% and AR50:95 of 11.5%), it performs relatively well in detecting Tuta absoluta and leaf miner, despite the underrepresentation of these classes in the dataset.

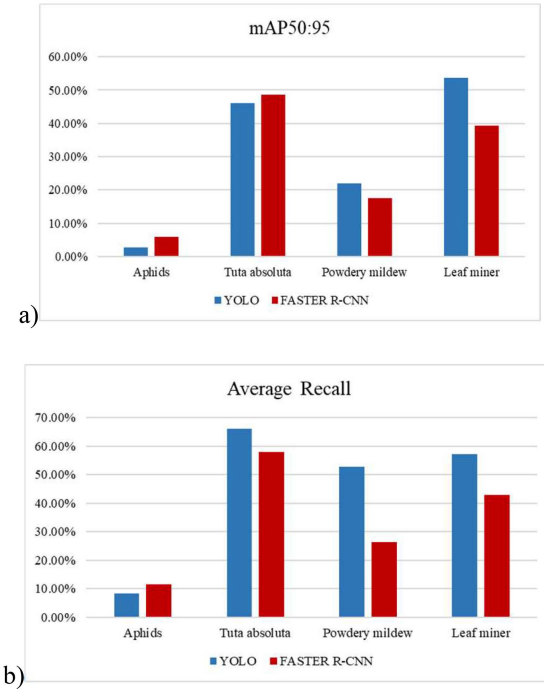

Fig. 9. Comparison of YOLO and Faster R-CNN. (a) mAP50:95. (b) AR.

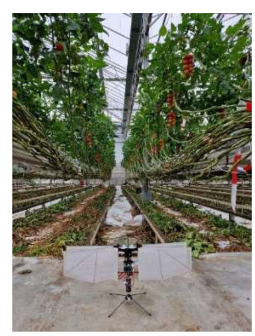

Fig. 10. FWD in greenhouse.

G. Models Comparison: YOLO Versus Faster R-CNN

Upon comparing the results of both models, it becomes evident that despite the Faster R-CNN being touted as a more accurate two-stage object detection algorithm, the YOLO algorithm outperformed it in our scenario, achieving higher mAP50:95 and AR scores in total and per class (see Fig. 9). This is notable even after providing the Faster R-CNN with input images of larger size. In both models, the most challenging class was aphids, with both models struggling to accurately detect them.

H. DL Models Tested on Data Acquires By the FWD

This section demonstrates the capability of the developed DL models to process and analyze data (videos/images) acquired by an FWD. The drone was deployed in our greenhouse compartment at the World Horti Center in Naaldwijk, The Netherlands (see Fig. 10). The video was captured when powdery mildew was predominantly affecting the plants in the compartment.

For the purpose of evaluating the DL model performance on the FWD video, annotations were generated at a rate of 7

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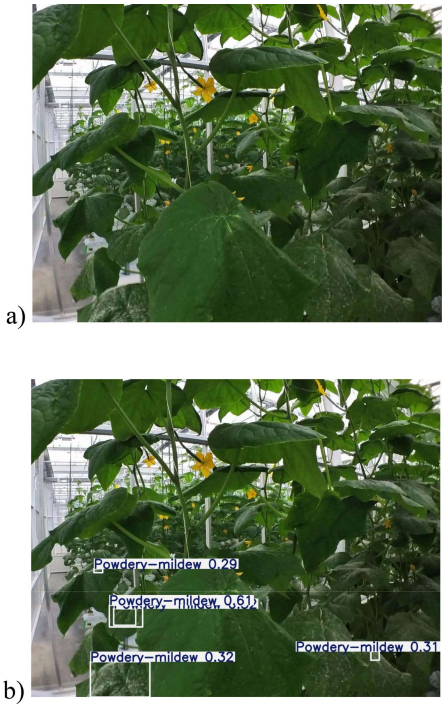

Fig. 11. (a) FWD video frame and (b) corresponding frame analyzed by YOLO.

frames/s, resulting in a total of 116 annotated frames. Fig. 11(a) and (b) presents a video frame and the corresponding frame analyzed by the YOLO DL algorithm, respectively. The results indicate that the DL model can detect powdery mildew with an accuracy ranging from 0.29 to 0.61. In addition, the DL algorithm is capable of detecting powdery mildew at varying distances and depths.

From the tested video, it was observed that the YOLO model detected the powdery mildew relatively well, considering the shaking movement in the video caused by the flapping wings of the drone. The model achieved a precision of 22% with an IoU threshold of 0.3. The performance of the Faster R-CNN was worse, as only a few predictions appeared to match the true annotations.

V. Concluding Remarks and Future Research

This study trained, tested, and validated two DL algorithms, YOLO and Faster R-CNN. The global performance metrics, mAP50 and mAP50:95, were 50% and 31% for YOLO, and 28% and 34.7% for Faster R-CNN, respectively.

Despite being underrepresented in the current dataset, our DL models demonstrated the ability to detect leaf miner and Tuta absoluta with the highest mAP (mAP50). This is attributed to their relatively large size and distinct structural features, which facilitate easier identification compared to smaller pests such as aphids.

Videos acquired by the FWD were analyzed using the two DL algorithms (YOLO and Faster R-CNN). YOLO exhibited superior performance, achieving a precision of 27.50 with an IoU of 0.4.

Future research should focus on extending the current dataset by collecting data from various types of crops and different pests and diseases. The DL models will be further developed to not only recognize pests but also provide information about pest and disease evolution and life cycles. In addition, within the framework of the Flappy project, efforts are being made to enhance performance through various hardware and software advancements.

Improvements in pest detection accuracy can be achieved through the acquisition of higher resolution images. One approach to attaining this objective is to reduce the size of the FWD, thereby enabling closer proximity to foliage for image capture. A dedicated work package within the Flappy project focuses on redesigning the actuation mechanism by drawing inspiration from insect physiology. This approach aims to increase the flapping frequency while reducing the wingspan [32], thereby enhancing the drone maneuverability and imaging capabilities.

Furthermore, efforts are underway to develop an alternative power supply system to replace conventional batteries. Extending flight duration and increasing payload capacity are critical considerations, as they would enable the integration of additional sensors for autonomous navigation and pest detection. Research is currently being conducted on alternative energy sources, such as chemical fuels (e.g., hydrogen peroxide), which offer higher power density than conventional batteries.

On the software side, efforts are focused on improving the performance of DL algorithms for pest detection. In addition to YOLO and the Faster R-CNN, other models will be explored to determine whether specific algorithms demonstrate superior performance for detecting particular pests. For example, preliminary investigations may reveal that YOLO is more effective in detecting aphids, whereas the Fast R-CNN performs better in identifying thrips. Given that real-time processing is a fundamental requirement of this work, optimization strategies have been employed to minimize computational time. However, to further enhance detection accuracy, preprocessing techniques will be introduced to extract regions of interest from images and video frames that are indicative of pest presence. In addition, hyperparameter optimization will be performed to refine the performance of DL models.

Pest and disease monitoring systems based on FWDs have the potential to play a crucial role in the IPM [33]. Monitoring, a fundamental component of the IPM, is traditionally conducted using yellow sticky traps, which do not facilitate precise pest localization and quantification within greenhouses. Furthermore, this activity is often performed manually, which presents challenges in light of the increasing labor shortages affecting the Dutch horticulture sector. Future developments within the Flappy project will focus on extracting key parameters from images and videos, such as pest density, which serve as essential inputs for risk analysis models. These models can correlate pest severity—quantified, for instance, as the number of insects per unit area or per leaf—with plant health and productivity.

Such models offer valuable support to growers in the decision-making process, an integral aspect of the IPM. By incorporating information on pest type, life cycle, distribution, and localization, risk analysis models can estimate financial losses and

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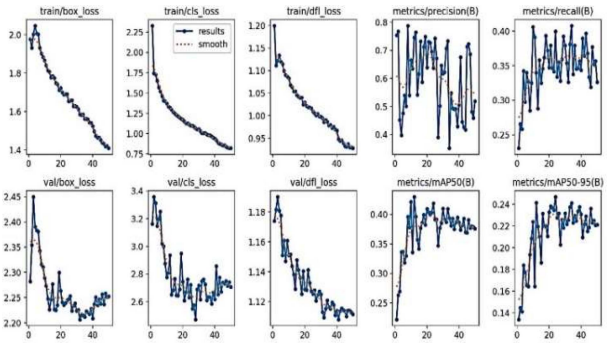

Fig. 12. Loss and accuracy plots: YOLO.

potential damage, thereby guiding growers in determining whether intervention is necessary and, if so, selecting the most effective control strategy. In the IPM, chemical treatments are typically considered a last resort, as growers increasingly adopt biological control measures. Therefore, it is imperative to develop DL algorithms capable of detecting not only pests but also their natural predators. Monitoring the balance between pest populations and predatory species is essential for the effective implementation of biological control strategies, and FWDs can play a pivotal role in facilitating this process. In the Dutch horticultural landscape, greenhouses are continuously increasing in size. Within the Flappy project, we aim to address the scalability challenges of FWD-based monitoring systems by developing and training swarms of FWDs.

Appendix I
YOLOv9 Architecture

Our model was deployed with the backbone CSPDarknet53 due to its alignment with our computational resources and its satisfactory precision score. We employed the pretrained weights of YOLOv9-C and further trained the model on our dataset. The initial ten convolutional layers were frozen to preserve their learned features, while the remaining layers (608 layers in total) were fine-tuned to enhance feature representations pertinent to our task of recognizing pests and diseases.

Appendix II
Loss and Accuracy Plots of the Best YOLOv9 Trained Model

A review of the graphs (see Fig. 12) indicates that, using the optimal set of hyperparameters, the validation mAP50 and mAP50-95 matrices obtain the highest values around the 20th epoch. The YOLO model's training process demonstrates effective learning, with training losses (box, classification, and objectness) consistently decreasing and metrics such as precision, recall, and mAP improving steadily, reflecting better detection performance. However, the increase in validation classification loss and box loss toward the end of training suggests potential overfitting, where the model becomes too specialized to the training data, losing its ability to generalize well to unseen data. These trends highlight the need for measures such as early stopping or regularization to enhance stability and generalization.

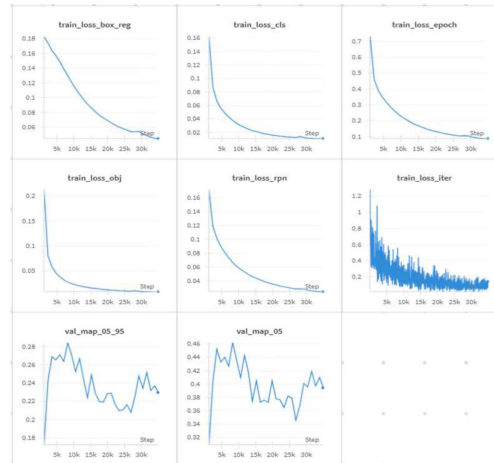

Fig. 13. Visualization best hyperparameters results of the Faster R-CNN.

Appendix III
Faster R-CNN Architecture

We have chosen to train the Faster R-CNN algorithm with the ResNet50 FPN V2 backbone, which is characterized by its improved ability to detect small objects [34].

Appendix IV
Loss and Accuracy Plots of the Best Faster R-CNN Trained Model

These graphs illustrate the model performance on both the training and validation datasets (see Fig. 13), aiding in monitoring the training progress and identifying potential overfitting or underfitting issues. They help assess whether the model has been trained effectively and can accurately recognize the features of our objects. The training loss graphs show consistent declines across all components (classification, bounding box regression, objectness, and RPN loss), indicating effective learning and convergence without signs of overfitting. The validation metrics, val_map_0.5 and val_map_0.5:0.95, exhibit variability but trend toward stabilization, reflecting the model improving detection performance across different IoU thresholds. While the results suggest robust learning and generalization, the fluctuations in validation metrics may require further tuning or validation dataset analysis to enhance consistency.

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